Connected to Python 3.13.14

Smart Fitness Session Analyzer Report To run the file Install the VS Code Python and Jupyter extensions and open this file in VS Code. and the ipykernel package must be installed beforehand.

Each section below demonstrates one bullet point from the assignment's "Common requirements" list, in the same order they are listed there.

1. At least four meaningful classes, including one clear example of composition

In [ ]:
```
from src.observation import Observation
from src.participant import Participant
from src.session import Session
from src.classifier import SessionClassifier

participant = Participant("John Doe", 70, 0.5)
session = Session(participant)  # composition: a Session has a Participant and has

observation_data = {
    "timestamp": 1,
    "heart_rate": 70,
    "skin_response": 0.5,
    "temperature": 36.6,
    "activity_level": 0.8,
    "signal_quality": 0.9,
}
success, message = session.add_observation(observation_data)
assert success

print("Classes:", Participant.__name__, Observation.__name__, Session.__name__, Ses
print("Composition check:", session.participant is participant, isinstance(session.
```

```
Setting resting heart rate to 70
Setting resting activity level to 0.5
Classes: Participant Observation Session SessionClassifier
Composition check: True True
```

2. At least one private or protected-style attribute controlled through a property

In [ ]:
```
print("Protected attribute:", participant._resting_heart_rate)

try:
    participant.resting_heart_rate = 999
except ValueError as error:
    print("Invalid assignment correctly rejected:", error)
```

```
Protected attribute: 70
Invalid assignment correctly rejected: Resting heart rate must be between 30 and 12
0.
```

3. Composition preferred over inheritance

`Session` is not a type of `Participant` or `Observation` - it aggregates them. No inheritance hierarchy is used between the domain classes for this reason

4. At least one class method or static method with a justified purpose

In [ ]:
```
result = SessionClassifier.classify_session(session)  # static method, needs no ins
print("classify_session (static method):", result["classification"])
print(SessionClassifier.get_classification_rules())  # class method, documents ACTI
```

```
classify_session (static method): insufficient data
Classification rules:
  resting: heart rate increase below 10%
  moderate activity: heart rate increase below 55%
  high activity: heart rate increase above the moderate threshold
  recovering: heart rate and activity decline near the end of the session
  insufficient data: fewer than 3 usable observations
```

5. At least four standalone functions for calculations, validation, or presentation

In [ ]:
```
from src.calculations import calculate_average, calculate_min, calculate_max, calcu
from src.validation import validate_observation, is_good_signal_quality
from src.report import generate_session_report, print_session_report

heart_rates = [70, 75, 80, 65, 90]
print("calculate_average:", calculate_average(heart_rates))
print("calculate_min:", calculate_min(heart_rates))
print("calculate_max:", calculate_max(heart_rates))
print("calculate_hr_elevation:", calculate_hr_elevation(90, 70))
print("validate_observation:", validate_observation(observation_data))
print("is_good_signal_quality:", is_good_signal_quality(0.9))
```

```
calculate_average: 76.0
calculate_min: 65
calculate_max: 90
calculate_hr_elevation: 20
validate_observation: (True, '')
is_good_signal_quality: True
```

6. Lists and dictionaries as part of the application data

In [ ]:
```
summary = session.get_session_summary()
print("Session summary (dict):", summary)
print("Observations (list):", session.get_observations())
```

```
Session summary (dict): {'total_observations': 1, 'good_quality_observations': 1, 'a
verage_heart_rate': 70.0, 'min_heart_rate': 70, 'max_heart_rate': 70, 'average_skin_
response': 0.5, 'average_activity_level': 0.8, 'average_temperature': 36.6}
Observations (list): [<src.observation.Observation object at 0x0000021A60F55E80>]
```

7. Validation of missing, impossible, or poor-quality values

In [ ]:
```
missing_field = {k: v for k, v in observation_data.items() if k != "heart_rate"}
impossible_value = {**observation_data, "heart_rate": 999}
poor_quality = {**observation_data, "signal_quality": 0.2}

print("Missing field:", validate_observation(missing_field))
print("Impossible value:", validate_observation(impossible_value))
print("Poor signal quality accepted by validation, but excluded from analysis:", is
```

```
Missing field: (False, 'Missing required key: heart_rate')
Impossible value: (False, 'Heart rate must be between 30 and 230.')
Poor signal quality accepted by validation, but excluded from analysis: False
```

8. A structured result returned as a dictionary

In [ ]:
```
report_dict = generate_session_report(session, result)
print(report_dict)
```

```
{'participant': 'John Doe', 'resting_heart_rate': 70, 'classification': 'insufficien
t data', 'confidence': 0.0, 'reason': 'Only 1 usable observations; at least 3 are re
quired.', 'total_observations': 1, 'usable_observations': 1, 'usable_percent': 100.
0, 'avg_heart_rate': 70.0, 'min_heart_rate': 70, 'max_heart_rate': 70, 'avg_activity
_level': 0.8}
```

9. A readable console report

In [ ]:
```
print_session_report(session, result)
```

```
============================================================
FITNESS SESSION ANALYSIS REPORT
============================================================
Participant: John Doe
Resting heart rate: 70 bpm
------------------------------------------------------------
Classification: INSUFFICIENT DATA (confidence 0%)
Reason: Only 1 usable observations; at least 3 are required.
------------------------------------------------------------
Observations: 1/1 usable (100.0%)
Heart rate: avg 70.0, min 70, max 70
Average activity level: 0.80
============================================================
```

10. At least five scenarios, including normal, unusual, and invalid-data cases

In [ ]:
```
from sample_data import (
    get_resting_session_data,
    get_moderate_activity_session_data,
    get_high_activity_session_data,
    get_activity_followed_by_recovery_session_data,
    get_poor_quality_session_data,
)


def build_session(participant_data, observations_data):
    """Create a Session and add all observations, printing any that get rejected."""
    p = Participant(
        participant_data["name"],
        participant_data["resting_heart_rate"],
        participant_data["resting_activity_level"],
    )
    s = Session(p)
    for observation in observations_data:
        ok, msg = s.add_observation(observation)
        if not ok:
            print(f"  Rejected: {msg}")
    return s


scenarios = [
    ("Resting session", get_resting_session_data),
    ("Moderate activity", get_moderate_activity_session_data),
    ("High activity", get_high_activity_session_data),
    ("Activity followed by recovery", get_activity_followed_by_recovery_session_dat
    ("Poor quality / invalid sensor data", get_poor_quality_session_data),
]

for label, data_fn in scenarios:
    print(f"\n--- {label} ---")
    p_data, o_data = data_fn()
    s = build_session(p_data, o_data)
    print_session_report(s, SessionClassifier.classify_session(s))
```

```
--- Resting session ---
Setting resting heart rate to 78
Setting resting activity level to 0.05
============================================================
FITNESS SESSION ANALYSIS REPORT
============================================================
Participant: Ola Normann
Resting heart rate: 78 bpm
------------------------------------------------------------
Classification: RESTING (confidence 100%)
Reason: Average heart rate stays close to the resting baseline.
------------------------------------------------------------
Observations: 18/18 usable (100.0%)
Heart rate: avg 79.3, min 76, max 85
Average activity level: 0.06
============================================================

--- Moderate activity ---
Setting resting heart rate to 59
Setting resting activity level to 0.1
============================================================
FITNESS SESSION ANALYSIS REPORT
============================================================
Participant: Kari Normann
Resting heart rate: 59 bpm
------------------------------------------------------------
Classification: MODERATE ACTIVITY (confidence 100%)
Reason: Average heart rate is moderately elevated above baseline.
------------------------------------------------------------
Observations: 18/18 usable (100.0%)
Heart rate: avg 81.8, min 74, max 98
Average activity level: 0.41
============================================================

--- High activity ---
Setting resting heart rate to 71
Setting resting activity level to 0.08
============================================================
FITNESS SESSION ANALYSIS REPORT
============================================================
Participant: Per Normann
Resting heart rate: 71 bpm
------------------------------------------------------------
Classification: HIGH ACTIVITY (confidence 100%)
Reason: Average heart rate is significantly elevated above baseline.
------------------------------------------------------------
Observations: 17/17 usable (100.0%)
Heart rate: avg 145.3, min 116, max 161
Average activity level: 0.87
============================================================

--- Activity followed by recovery ---
Setting resting heart rate to 60
Setting resting activity level to 0.06
============================================================
FITNESS SESSION ANALYSIS REPORT
```

```
============================================================
Participant: Eva Normann
Resting heart rate: 60 bpm
------------------------------------------------------------
Classification: RECOVERING (confidence 100%)
Reason: Heart rate and activity decline toward the end of the session.
------------------------------------------------------------
Observations: 17/17 usable (100.0%)
Heart rate: avg 88.5, min 61, max 125
Average activity level: 0.44
============================================================

--- Poor quality / invalid sensor data ---
Setting resting heart rate to 66
Setting resting activity level to 0.1
  Rejected: heart_rate must be numeric.
  Rejected: Heart rate must be between 30 and 230.
  Rejected: Activity level must be between 0.0 and 1.0.
  Rejected: skin_response must be numeric.
  Rejected: heart_rate must be numeric.
  Rejected: Heart rate must be between 30 and 230.
  Rejected: Activity level must be between 0.0 and 1.0.
  Rejected: Heart rate must be between 30 and 230.
  Rejected: Temperature must be between 25 and 42.
  Rejected: Activity level must be between 0.0 and 1.0.
  Rejected: Heart rate must be between 30 and 230.
  Rejected: Temperature must be between 25 and 42.
============================================================
FITNESS SESSION ANALYSIS REPORT
============================================================
Participant: Nils Normann
Resting heart rate: 66 bpm
------------------------------------------------------------
Classification: MODERATE ACTIVITY (confidence 80%)
Reason: Average heart rate is moderately elevated above baseline.
------------------------------------------------------------
Observations: 4/5 usable (80.0%)
Heart rate: avg 90.0, min 87, max 92
Average activity level: 0.50
============================================================
```

Test coverage

Runs the actual test suite in `tests.py`.

In [ ]:
```python
import runpy

_ = runpy.run_path("tests.py", run_name="__main__")
```

```
Setting resting heart rate to 60
Setting resting activity level to 0.5
Setting resting heart rate to 70
Setting resting activity level to 0.8
PASS  test_participant
Setting resting heart rate to 60
Setting resting activity level to 1
PASS  test_participant_invalid_values
PASS  test_observation
PASS  test_validation
PASS  test_calculations
Setting resting heart rate to 70
Setting resting activity level to 0.5
PASS  test_session
Setting resting heart rate to 70
Setting resting activity level to 0.5
PASS  test_session_rejects_invalid_observation
Setting resting heart rate to 60
Setting resting activity level to 0.1
Setting resting heart rate to 60
Setting resting activity level to 0.1
Setting resting heart rate to 60
Setting resting activity level to 0.1
PASS  test_classifier
Setting resting heart rate to 60
Setting resting activity level to 0.1
PASS  test_report

9/9 tests passed
```